

RateEdge Options Platform

Professional Interest Rate Derivatives Pricing

Platform Overview & Capability Summary | v28 | March 2026

Built & Designed By

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Executive Summary

RateEdge Options Platform is a cloud-hosted, professional-grade interest rate derivatives pricing system purpose-built for the Australian and G10 rates markets. It brings institutional-quality pricing mathematics — the frameworks used by global investment banks and interdealer brokers — to a clean, browser-based interface designed for practitioners who need answers in seconds.

The platform spans the full rates derivatives spectrum: vanilla IRS, swaptions, caps and floors, zero coupon swaps, OTC bond options, curve and spread options, CMS products, Bermudan callables, and an integrated relative value engine. All products share a consistent dual-curve framework — IRS curves for rate projection, OIS/AONIA for discount factors — and the AFMA-compliant Australian business day calendar with Modified Following throughout.

Its distinguishing characteristic is the depth of market knowledge embedded in every formula. The platform was not assembled from textbooks — it reflects twenty years of front-line experience pricing, broking, and risk-managing these instruments across New York, Hong Kong, and Sydney. The difference between a formula that works in theory and one that matches a bank's live screen almost always comes down to market convention, curve construction, or settlement detail. Those details are only learned by doing.

Built on Two Decades of Institutional Rates Experience

The pricing logic, market conventions, and product scope of RateEdge were defined by someone who has traded and brokered these instruments in live interdealer markets for over twenty years — across the most technically demanding rates markets in the world.

- Began in Sydney in 1992, broking forward foreign exchange thirds at Babcock Fulton Prebon.
- Moved to Hong Kong and started in Swaps on USD and Yen at Tullett Hong Kong in 1994, moving to Tokyo Forex in 1996 to work on the Yen IRS desk as the Global Linkman.
- Started at Yamane Prebon on the Yen IRS desk, moving to the Options desk in 1998.
- USD interest rate options at Cantor Fitzgerald in New York in 2003, then moved to BGC Partners where five years were spent brokering USD IRO — including three years as Regional Manager for interest rate options across Asia-Pacific from Hong Kong.
- At GFI Group over nearly a decade (Hong Kong then Sydney) ran AUD interest rate options as Manager and Director, developing deep expertise in BBSW basis structure, Q/Q vs S/S coupon conventions, AFMA settlement, and the unique dynamics of the Australian rates curve.
- As Manager IRO at ICAP Sydney and subsequently as Chairman & CEO of PO Capital Markets, built electronic trading infrastructure, OMS platforms, and analytics tools for the interdealer rates market — developing a precise understanding of what practitioners actually need from a pricing system.
- Founded RateEdge in 2025 to bring that accumulated expertise together in a single, accessible platform — one that prices correctly, communicates clearly, and generates client-ready output.

This background is directly visible in the platform's technical choices: AFMA Modified Following conventions in all date calculations, correct Q/Q vs S/S frequency switching by tenor, BBSW 3m projection with OIS discounting, the AOFM dirty price formula for AGB bond options, Hagan LSR convexity adjustment for CMS products, and Jamshidian decomposition for Bermudan calibration. These are not approximations — they are the market standard.

Platform Architecture & Access

Module / Feature	Capability
Hosting	Microsoft Azure — browser-accessible, no installation required. Works on desktop, tablet, and mobile.
Authentication	Email OTP — secure, no password management. Full audit trail.
Currencies	AUD (full), NZD, USD, (EUR, GBP, JPY, CAD under development) — IRS and derivatives support.
Settlement	Full AFMA Conventions Australian business day calendar — NSW holidays, Modified Following throughout all date rolls.
Dual-Curve	IRS curves for rate projection; OIS/AONIA for annuity discounting. Consistent across every product.
Data Storage	Azure PostgreSQL for vol snapshots. Local JSON persistence for conventions, spreads, blotter, correlation matrix.

Output	Branded Excel client schedules, vol surface exports via email (Office 365/Resend), CSV portfolio download.
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IRS Pricer – Separate Licence

Multi-currency IRS pricing with full bucket DV01 breakdown, AFMA \$25k DV01 parcel lot calculations, and YM/XM bond futures EFP hedging calculator. Correct Q/Q ($\leq 3Y$) vs S/S ($> 3Y$) leg frequency for AUD with 6v3 basis curve adjustment. 22-expiry support. Currency overview page with IRS market convention summaries.

Vol Surface, SABR & Rate/Vol Matrix

- Upload or manually configure the full ATM vol surface — swaption and caplet matrices separately.
- SABR parameter matrices (α , β , ρ , v) configurable per expiry/tenor point. Implied vol surface generated on demand.
- 3D drag-and-drop vol editor with real-time surface recalculation and Apply/Confirm workflow.
- Vol history snapshots saved to database with timestamps — compare surfaces across dates.
- Vol export tab: generate formatted Excel workbooks and distribute by email.
- Rate/Vol Matrix: forward swap rates (22×12), ATM premium matrix, Vega matrix (\$/1bp per 100mm notional). Leg convention toggle. Four decimal display.

Swaptions Pricer

Full swaption pricer supporting Normal (Bachelier) and Black (lognormal) models across the complete AUD, NZD, and USD vol surfaces.

- Expiries from 1 week to 20 years, with custom calendar date entry.
- Swap tenors 1Y to 30Y. Leg convention control (Market / Q/Q / S/S).
- Structures: Payer, Receiver, ATM Straddle, Strangle, Risk Reversal, Payer Ladder, Receiver Ladder. Defaults to ATM Straddle.
- CFS (Cumulative Forward Spread) premium mode alongside vanilla Fwd/Spot. Definitive wedge formulas: CFS bid = swaption bid + wedge bid; CFS offer = swaption offer + wedge offer.
- Forward pulled directly from the Rate/Vol Matrix when loaded — guarantees exact premium match to the matrix.
- Delta, Gamma, Vega, Theta, BPV returned. Persistent blotter with expiry-aware sort (1m, 2m, 3m → 1y, 2y...) and one-click reload to pricer.

Caps & Floors

Cap, Floor, Straddle, Collar, and Strangle structures. Caplet vol term structure bootstrap with cumulative forward spread (CFS) premium targets. Full caplet schedule with per-caplet DV01. Wedge calculation for FWD CFS straddles. Volatility curve chart per currency. Swaption and caplet vol surfaces treated independently, consistent with market practice.

Zero Coupon Swaps

Full ZCS pricer using **ending notional** (maturity amount) as the primary input — consistent with global market convention. Starting notional is back-calculated from the fixed rate. Fixed payment = $N_{\text{end}} - N_{\text{start}}$. Floating leg: quarterly or semi-annual BBSW resets compound over life, single payment at maturity.

- AFMA calendar-aware reset schedule — actual business day dates, opening/closing balances per period, period interest, cumulative accrued interest.
- Effective date = T+1 AUD business day (AFMA). Maturity date = Modified Following anniversary.
- Generates a branded Excel confirmation with three worksheets: Trade Summary (Fixed Rate Payer / Floating Rate Payer with LEI & BIC, LCH clearing details, ASIC reporting note), Floating Leg Reset Schedule, and Fixed Leg & Final Exchange.
- LCH SwapClear eligible (AUD BBSW, up to 31Y). CFTC-mandated clearing 28 days to 30 years residual maturity.

Bond Options — OTC Physical AGB

European options on Australian Government Bonds priced using the full AOFM/RBA dirty price formula with actual coupon schedules. Forward dirty price via GC repo carry with coupon stripping discounted at OIS. Black-76 (price lognormal) and Bachelier (yield normal) models. Strike entry in yield terms, mapped to price via AOFM formula at expiry. Delta, Gamma, Vega, Theta in both price and yield space. All 17 current AGB lines pre-loaded (coupon, maturity, settlement).

Exotics

Curve / Spread Options

Options on the spread between any two swap rates — steepener (payer), flattener (receiver), straddle. **Spread vol** = $\sqrt{(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)}$. Both leg vols drawn from the loaded ATM surface with expiry interpolation. Correlation defaults from a configurable 8x8 matrix (2Y–30Y). Defaults to Straddle. Payoff chart and correlation sensitivity analysis included.

CMS Products

Constant Maturity Swap caplets, floorlets, full cap/floor series, CMS swap fair rate, and CMS spread cap/floor. **Hagan Linear Swap Rate convexity adjustment** ($\text{adj} \approx F^2 \sigma^2 T \cdot |dA/dF|/A$) with AUD Q/Q vs S/S annuity convention. User-configurable per-tenor convexity bumps. Full caplet schedule with per-period convexity adjustment, adjusted forward, and PV.

Bermudan Swaptions & Callable Swaps

Hull-White 1F trinomial tree calibrated to co-terminal swaptions from the loaded ATM surface using Jamshidian decomposition. Backward induction compares exercise vs continuation at each node. Reports Bermudan PV, first-call European PV, and the Bermudan premium over European. Callable swap = Bermudan + underlying swap.

Correlation & Convexity Configuration

Persistent configuration panel at the top of the Exotics tab — editable 8×8 swap rate correlation matrix (2Y–30Y) and per-tenor CMS convexity bumps. Flows automatically into all spread option and CMS pricing. Saves to local JSON across sessions.

Relative Value Engine

The RV tab provides four integrated analysis tools combining the live vol surface and IRS curve with historical AUD rate data:

- **ATM vol vs calendar wings across all tenors — heatmapped richness/cheapness. Calendar spread ratios. Vol cone percentile positioning vs 3-year history.** Vol Butterfly Matrix:
- **Live key spreads with historical time-series, percentile ranking, and median vs current positioning. Forward spread vs curve spread scatter with linear regression, R^2 , and residual time-series — directly replicating the interdealer RV workflow.** Curve RV & Spread Analysis:
- **Scored and ranked recommendations: vol butterfly richness/cheapness, calendar spread opportunities, curve-driven directional swaptions (payer/receiver based on forward curve shape), gamma/theta relative value signals.** Swaption Trade Ideas:
- **Forward BBSW path visualisation from the curve. Curve-slope-driven cap vs floor preference. OTM cap recommendations at the implied forward rate peak. Vol cone cheap/rich entry signals.** Cap/Floor Trade Ideas:

Portfolio Blotter & Workflow

- Persistent swaption blotter — tickets survive page refresh, written to local JSON.
- Expiry-aware sort: 1m, 2m, 3m → 1y, 2y... (not alphabetical).
- Reload & Reprice: one-click restore of any blotter ticket back into the swaption pricer with all inputs pre-populated.
- Portfolio tab aggregates swaptions and caps/floors with full Greeks table, CSV download, and row removal.

Key Differentiators

	RateEdge Options Platform	Generic Pricer / Bloomberg
Conventions	Full AFMA — Modified Following, Q/Q vs S/S, BBSW basis, AOFM bond formula, AFMA holiday calendar	Approximate or generic
Dual-curve	IRS projection + OIS discounting throughout, consistent across all products	Varies; often single curve
Product depth	IRS · Swaptions · Caps/Floors · ZCS · AGB Bond Options · CMS · Bermudans · Spread Options · RV Engine	Typically, 2–3 product types
Domain knowledge	Built by a 20+ year IRO/IRS practitioner — formulas replicate live market screens, not textbook approximations	Generic financial engineering
Customisation	Vol surface editor · SABR config · Correlation matrix · CMS convexity overrides — all user-controlled	Fixed model parameters
Client deliverables	Branded Excel schedules with LEI/BIC, AFMA-accurate dates, LCH clearing & ASIC reporting notes	Usually none
Cost	SaaS subscription per user USD 1250 pm (USD 15000 pa) – after initial 2 licences 50% discount.	Bloomberg: ~USD 30,000 pa

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